

Nucleic acids

Nucleic acid was first isolated in 1868 by a swiss physician Friedrich Miescher from the nuclei of pus cell.

He named it as Nuclein.

Nuclein was renamed as Nucleic acid by Altman in 1889.

Nucleic acids are strong acid and are present in nucleus, mitochondria, plastids and in cytoplasm.

They occur in bacteria and in viruses.

Composition

The nucleic acids are polymers of nucleotides, or they are nucleotide polymers.

The nucleotides consists of

- i. Pentose sugar
- ii. Nitrogenous bases
- iii. Phosphate component.

A nucleoside is formed by combination of pentose sugar and nitrogenous base.

Nucleoside= Pentose sugar +nitrogenous base

Nucleotide= Pentose sugar +nitrogenous base+ phosphate

A. Phosphoric Acid

It contains a phosphate group and present in outermost part of nucleotide.

B. Pentose sugar(5- carbon sugar): These are two types of pentose sugar:

i. Ribose sugar

It is found in ribonucleic acid. One oxygen molecule is present at second carbon of pentose sugar.

ii. Deoxyribose sugar: It is found in deoxyribonucleic acid (DNA). One oxygen molecule is lacking at second carbon number of pentose sugar.

Nitrogenous bases

Nitrogenous bases are of two types

i. Purine

It is of two types

Adenine (A), Guanine (G)

ii. Pyrimidine

It is of two types

Thymine (T), Cytosine (C) and Uracil (U)

Complementary bases

The specific purine base is attached to the specific pyrimidine base with hydrogen bond. Adenine is attached with thymine by double hydrogen bond and cytosine attached with guanine with triple hydrogen bond.

Types

Nucleic acids are of two types

- A. Deoxyribonucleic acid (DNA)
- B. Ribonucleic acid (RNA)

Deoxyribonucleic acid (DNA)

DNA is a genetic material and forms molecular basis of heredity in all organisms except in TMV virus where RNA is genetic material.

In eukaryotic cells the DNA is in nucleus and is the main component of chromosomes, called as nuclear DNA.

A small amount of DNA is present in some cytoplasmic organelles like mitochondria and plastids, called as organellar DNA.

The DNA content is constant in all the cells of a given species. In interphase of cell division, the amount of DNA is doubled.

Chemical composition

DNA is the largest macromolecule in the organisms. It is a long double chain of deoxyribonucleotide units. The two chains are twisted around a common axis to form a right-handed double helix.

Each deoxyribonucleotide unit consist of three different molecules:

Phosphate, 5-carbon deoxyribose sugar and nitrogenous base.

Structure of DNA

Watson and Crick developed the double helical model of DNA.

The characteristics of DNA are as follows:

1. DNA consists of about 10^{10} deoxyribonucleotides. So, it is polydeoxyribonucleotide.
2. It is double stranded or double helically coiled structure which is spirally coiled around central axis in right-handed manner.
3. Two helices of DNA are antiparallel(one strand moves in 3'-5' direction but other strand moves in 5'-3' direction).
4. The diameter of DNA is 20 angstrom degree.
5. The distance of one complete spiral turn is 34 angstrom degree.
6. In one complete turn, there are 10 pairs of nitrogen bases. The distance between two pairs of nitrogen bases 3.4 angstrom degree.
7. Each deoxyribonucleotide consists of outer phosphoric acid, middle pentose sugar (deoxyribose sugar) and central nitrogenous base (A,T,C or G).

8. Nitrogen bases are of two types: purines (Adenine and Guanine) and pyrimidine (Cytosine and Thymine).

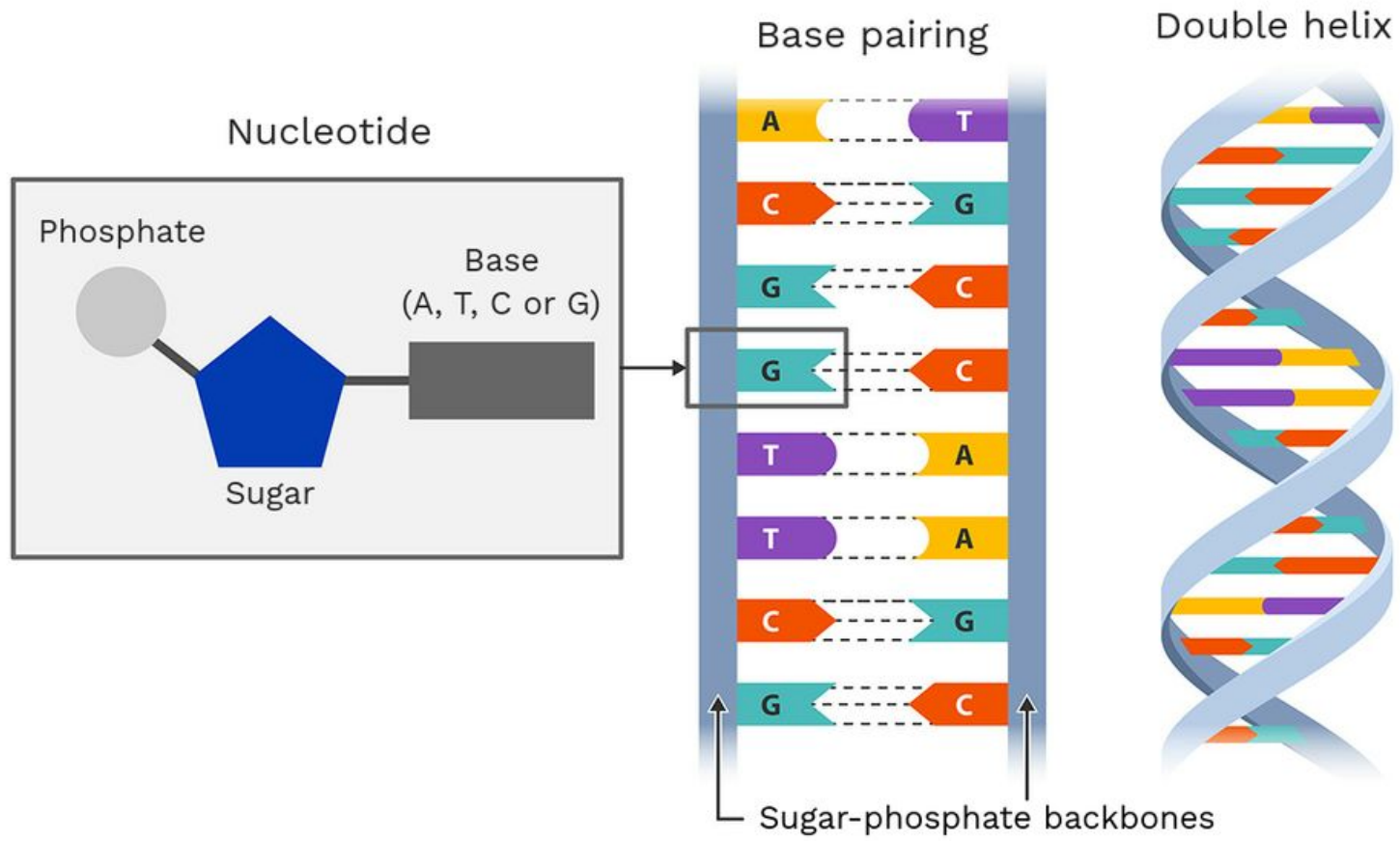
9. The complementary nitrogen bases of two strands of DNA are joined together by hydrogen bonds. Double hydrogen bond is present between Adenine and Thymine or vice versa ($A=T$ or $T=A$) and triple hydrogen bond is present between Cytosine and Guanine and vice versa ($C=G$ or $G=C$).

10. The phosphates and sugars of DNA helix are joined together by phosphodiester bond.

11. Glycosidic bond is present in between deoxyribose sugar and nitrogen base.

12. Each DNA has backbone of alternate deoxyribose sugar and phosphoric acid.

13. Nitrogen bases (A, T, C and G) are functional unit (information centre) of DNA.



Functions (biological roles) of DNA

1. DNA is a genetic material because it helps in the inheritance of traits in successive generations from parents to their offsprings.
2. Single parental DNA forms two offspring DNA by the process of DNA replication.
3. It synthesizes RNA by the process of Transcription.
4. It indirectly involves in protein synthesis and controls the metabolic activities of cell.
5. Developmental stages occur in the life cycle of an organism by an internal clock of DNA functioning.
6. Changes in sequence and number of nucleotides cause mutation which is a raw material of evolution of life.

7. DNA gives new combinations during meiosis by the process of crossing over.
8. It controls the development of individual through intrinsic and extrinsic flow of information.
9. It controls the growth and development of organisms.

Ribonucleic Acid (RNA)

Ribonucleic acid (RNA) is generally single stranded polynucleotide chain that mainly helps in protein synthesis.

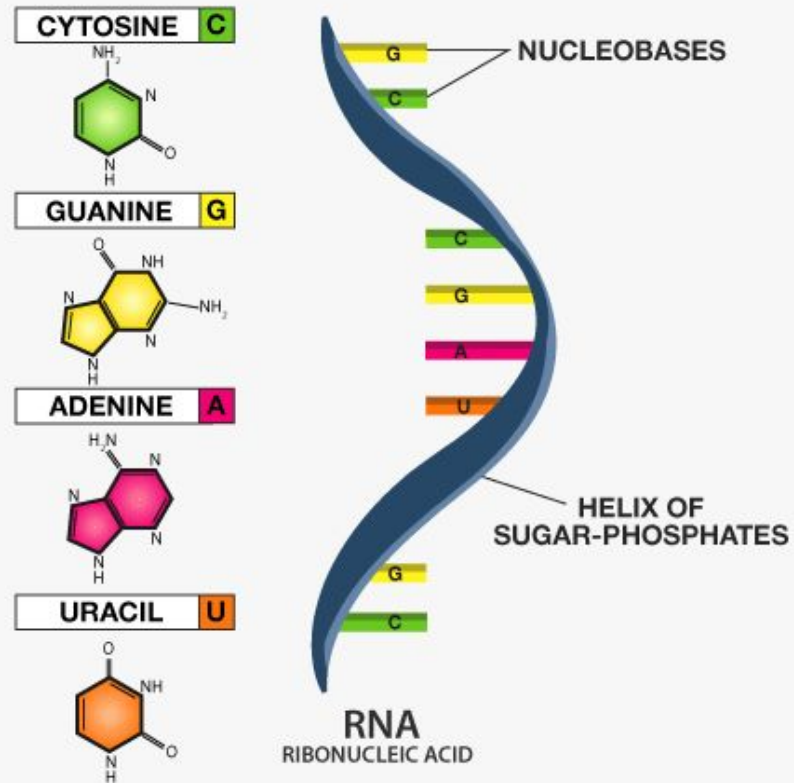
In some virus, it acts hereditary material . Virus as TMV , HIV etc have RNA as genetic materials.

It consists of 70-12000 nucleotides joined end to end by phosphodiester bond.

Each ribonucleotide consists of phosphate, pentose sugar(ribose sugar) and nitrogenous bases(A,U,C or G).

It is found in nucleolus , ribosomes, mitochondria, chloroplast ,nissl granules and cytoplasm.

RIBONUCLEIC ACID (RNA)



Types of RNA

Based on molecular size and functions, RNA is three types:

rRNA ,mRNA and tRNA

Ribosomal RNA (rRNA):

It is the most stable and abundant RNA comprising 70%-80% of total cellular RNA. It is associated with ribosome . It is synthesized from DNA in nucleolus by transcription in the presence of enzyme RNA polymerase-I.

mRNA (Messenger RNA)

It is a long RNA which constitutes 3%-10% of total cellular RNA.

mRNA is formed in nucleus with the help of DNA by process of transcription in the presence of enzyme RNA polymerase II.

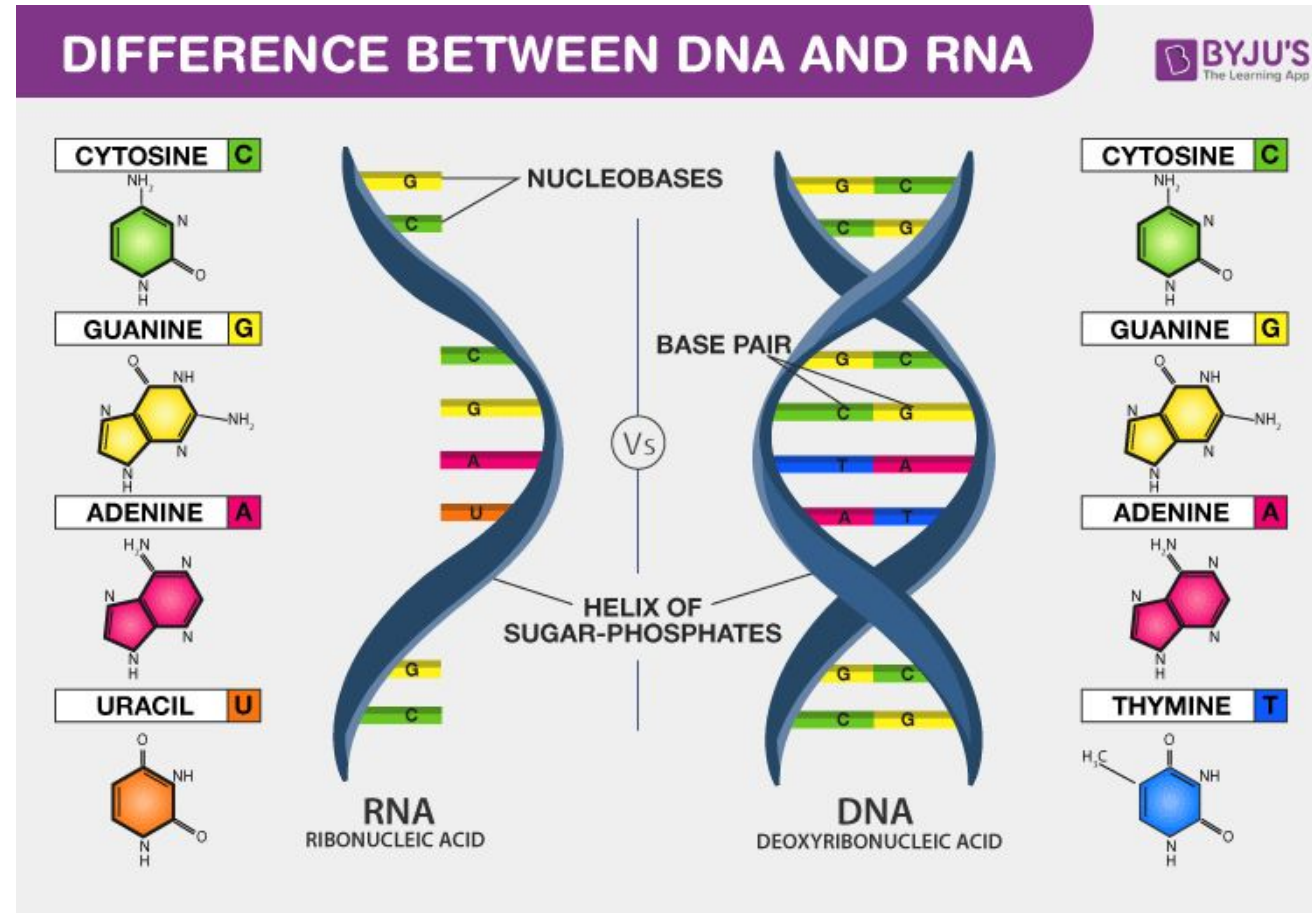
mRNA help in protein synthesis.

Transfer RNA (tRNA)

t RNA constitute about 10-20% of total cellular RNA . It is the smallest RNA with 70-85 nucleotides.

It is synthesized in nucleus from DNA by transcription in the presence of enzyme polymerase III.

Differences between DNA and RNA



DNA and RNA Difference

DNA (Deoxyribonucleic acid)	RNA (Ribonucleic acid)
Definition	
It is a long polymer. It has a deoxyribose and phosphate backbone having four distinct bases: thymine, adenine, cytosine and guanine.	Is a polymer with a ribose and phosphate backbone with four varying bases: uracil, cytosine, adenine and guanine.
Location	
It is located in the nucleus of a cell and in the mitochondria.	It is found in the cytoplasm, nucleus and in the ribosome.
Sugar portion	
It has 2-deoxyribose.	It has Ribose.
Function	
The function of DNA is the transmission of genetic information. It acts as a medium for long-term storage.	RNA is critical for the transmission of the genetic code that is necessary for protein creation from the nucleus to the ribosome.

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Predominant Structure	
DNA is a double-stranded molecule that has a long chain of nucleotides.	RNA is a single-stranded molecule which has shorter chain of nucleotides.
Propagation	
DNA replicates on its own, it is self-replicating.	RNA does not replicate on its own. It is synthesized from DNA when required.
Nitrogenous Bases and Pairing	
The base pairing is as follows: GC (Guanine pairs with Cytosine) A-T (Adenine pairs with Thymine).	The base pairing is as follows: GC (Guanine pairs with Cytosine) A-U (Adenine pairs with Uracil).